# Shor in Miniature & The RSA Reality Check

Hackathon Project: Quantum Cryptanalysis & Threat Modeling

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# Executive Summary: The Shor Pipeline

**Goal:** Implement a dynamic, mathematically rigorous quantum order-finding circuit (Part A) and quantify the hardware gap for breaking RSA-2048 (Part B).

- **Phase 1: Classical Pre-Checks** - Filtering prime guesses before invoking the quantum simulator.
- **Phase 2: Quantum Architecture** - Sizing registers and dynamically generating controlled permutation matrices.
- **Phase 3: QFT & Classical Decode** - Extracting frequency and bounding continued fractions.
- **Phase 4: The Reality Check** - Surface code mathematics proving a 2.5 million physical qubit requirement.

# Phase 1: Classical Pre-Checks

**The Concept:** Shor's quantum subroutine requires a random integer $a$ that is strictly coprime to the modulo $N$.

**Engineering Flex: Trap Avoidance**
If $gcd(a, N) > 1$, we accidentally guessed a factor. Pushing this to the quantum simulator causes fatal crashes. We filter it classically for an instant $O(1)$ win.

```
# Pick random 'a' between 2 and N-1
a = np.random.randint(2, N)

# Trap Check: Did we accidentally guess a factor?
gcd_a_N = math.gcd(a, N)
if gcd_a_N > 1:
    return gcd_a_N, N // gcd_a_N
```

# Phase 2: Register Initialization

**The Physics Concept:** We allocate $2n$ qubits to the Counting Register to achieve $2^{2n} \geq N^2$ precision.

**Engineering Flex: Strict State Initialization**
Applying Hadamard gates puts the counter in a massive superposition. Crucially, the Work Register must be initialized to $|1\rangle$, or modular multiplication evaluates to zero.

```
# 1. Superposition for parallel computation
for w in count_wires:
    qml.Hadamard(wires=w)

# 2. Trap Check: Initialize work register to |1>
# (PennyLane's least significant bit is the last wire)
qml.PauliX(wires=work_wires[-1])
```

# Phase 3: The Oracle

**The Goal:** Multiply the Work Register by $a^x \pmod{N}$ to sculpt the superpositions into a periodic wave.

**Engineering Flex: Dynamic Matrix Generation**
Instead of hardcoding permutation tables, we dynamically construct unitary matrices at runtime, safely mapping out-of-bounds states to the Identity to prevent overflow.

```
U = np.zeros((size, size))
for i in range(size):
    if i < N:
        target = (i * a_pow) % N
        U[target, i] = 1.0
    else:
        # Identity mapping for overflow states
        U[i, i] = 1.0

qml.ctrl(qml.QubitUnitary, control=ctrl_wire)(U, wires=work_wires)
```

# Phase 4: QFT & Classical Decode

**The Physics Concept:** The Inverse Quantum Fourier Transform (IQFT) forces destructive interference, extracting the hidden period $r$ into a single measured phase decimal.

**Engineering Flex: Bounding Continued Fractions**
Standard fraction algorithms will infinitely expand to match quantum simulator noise. We strictly bound the denominator to $N$.

```
# Convert binary measurement to float
phase_float = phase_int / (2**n_count)

# Trap Check: Bound the denominator to ignore quantum noise
frac = Fraction(phase_float).limit_denominator(N)
r = frac.denominator
```

# Part B: The Algorithmic Tradeoff

**The Question:** Why did estimates for factoring RSA-2048 drop from 20M physical qubits (2019) to under 1M qubits (2025), while runtime increased from 8 hours to 1 week?

**The Answer: Trading Space for Time**
Hardware capabilities did not change (both assume 0.1% gate error). Instead, researchers utilized algorithmic compression:

- **Yoked Surface Codes:** Packing logical qubits tighter.
- **Approximate Residue Arithmetic:** Reducing required bits.
- **Result:** Operations must be run sequentially rather than in parallel. The quantum computer becomes a slow, capacity-constrained mainframe.

# Part B: Surface Code Mathematics

**The Parameter Constraint:**
For $L = 1,500$ logical qubits running for 5 days ($T = 4.3 \times 10^{11}$ cycles), maintaining a $< 10\%$ failure rate requires:

$$L \cdot T \cdot p_L < 0.1 \implies p_L < 1.55 \times 10^{-16}$$

**Calculating Code Distance ($d$):**
Assuming physical error $p = 0.1\%$ and threshold $p_{th} = 1\%$:

$$10^{-16} \approx 0.1(0.1)^{\frac{d+1}{2}} \implies d = 29$$

**Total Hardware Requirement:**
Physical qubits per logical qubit $\approx 2d^2 = 1,682$.

$$\text{Total} \approx 1,500 \times 1,682 = \textbf{2.52 Million Physical Qubits}$$

# The Reality Check & Threat Model

**The Hardware Gap: 2,500x**
Current superconducting architectures (e.g., IBM Condor) possess roughly 1,000 qubits at 0.5% error rates. A 2.5 million qubit requirement proves cryptographic irrelevance for at least a decade.

**The HNDL Threat Model:**
If the hardware is decades away, why migrate now?

- **Harvest Now, Decrypt Later (HNDL):** Adversaries are currently recording encrypted TLS traffic.
- **The Imperative:** When the 2,500x gap closes, all data captured today will be retroactively decrypted. Migration to post-quantum standards (ML-KEM/Kyber) must begin immediately.

# Thank You

Result: A trap-resistant, dynamic Shor's architecture paired with mathematically proven cryptographic threat modeling.